1. A, B, C
2. We use the objective function for regression in a multilayered perceptron as

$$E(W, v|X) = -\sum_t [r^t log y^t + (1 - r^t) log(1 - y^t)]$$

We derive the update function for this parameter v by taking the gradient of the objective function with respect to v.

$$\Delta v_h = \eta \sum_t (r^t - y^t) z^t{}_h$$

3. An example of an activation function is the sigmoid activation function which takes the input and squeezes it to between 0 and 1. Typically the activation function outputs a 1 or a 0 which just conveys whether or not it was activated regardless of the magnitude of the input value. It activates based on the weights and inputs of one layer and becomes the input for the next layer in a multilayered perceptron.